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erlying health conditions and
psychological and environmental factors[3]. However, sleep disturbances can lead to adverse health outcomes, including heart disease, anxiety disorders, depression, falls, accidents, cognitive impairment, decreased quality of life, and even death[4].

Sleep consists of two stages: rapid eye movement (REM) sleep, characterized by such eye movements, intense brain activity and muscle relaxation; and non-rapid eye movement (NREM) sleep, which consists of three progressive stages ranging from light sleep to deep sleep. During NREM sleep, memory consolidation and physical recovery take place, while REM sleep plays a crucial role in emotional and cognitive processing, as well as being associated with vivid dreaming. These stages alternate throughout the night in successive cycles, promoting restorative, high-quality sleep[3].

Good sleep quality is characterized by the ability to maintain continuous and restorative sleep, which includes achieving an adequate deep-REM sleep proportion, experiencing minimal disruptions throughout the night and waking up feeling rested[5].

According to the Brazilian Sleep Association[6], older adults generally go to bed earlier than younger adults, enjoying more than seven consecutive hours of good-quality sleep. Daytime naps are common, while most of the sleep hours are at night. This pattern is typical and should not be regarded as a sleep disorder. However, factors such as low physical activity levels and social isolation (which are common during retirement) can affect sleep.

Poor sleep quality is one of the most common complaints among older adults and is a condition that can adversely affect health and individual well-being. Sleep is essential for promoting rest, physical and mental recovery, cognitive function and emotional well-being. Some studies suggest that poor sleep quality is associated with an increased risk of cognitive decline[4,7,8]. Aspects such as sleep duration, daytime sleepiness, naps and total sleep hours exert an influence on cognition and may lead to impaired cognitive performance in older adults[8].

The long-latency auditory evoked potential (P300) is an electrophysiological test that allows identifying potential dysfunctions and/or alterations in the central auditory nervous system (CANS) through an objective measure of cognitive processes, such as attention place approximately 300 milliseconds after the onset of a relevant stimulus for the task and can be elicited in response to visual, auditory or somatosensory stimuli[10]. Two variables are used to quantify P300: wave latency, which reflects the time required for information processing; and amplitude, which is associated with the attention level[11].

Few studies have investigated the relationship between sleep disorders and cognitive changes using the P300 test. One of them demonstrated that sleep deprivation caused by the obstructive sleep apnea syndrome (OSAS) leads to cognitive deterioration, as evidenced by changes in latency and amplitude of the P300 wave[12]. Another study, also related to OSAS, suggested using electrophysiological tests for cognitive assessment[13]. Both studies highlight the P300 wave as an essential tool for investigating cognitive changes related to sleep disorders.

It is believed that, by influencing cognitive performance, sleep quality may affect attention levels and the auditory information processing speed and, consequently, the P300 responses. The objective of this study was to examine the relationship between sleep quality and cognitive performance (P300) in older adults.

METHODS

This is a cross-sectional and observational study with a quantitative approach to the data. The project was approved by the Research Ethics Committee of the affiliated university (Certificate of Presentation for Ethical Appreciation—CAAE: 69137123.4.0000.5504; approval number: 6,482,515). All participants signed the free and informed consent form as required by Resolution No. 466/12 of the National Health Council. Upon agreement, a date and time for the assessments was scheduled, with all evaluations conducted between November 2023 and February 2024 at the Gerontology Department belonging to the Universidade Federal de São Carlos (UFSCar).

To carry out data collection, community-dwelling aged individuals were invited in person, either by close acquaintances or through referral. The study inclusion criterion was individuals aged at least 60 years old. People with bilateral hearing loss at 1,000 and 2,000 Hz at 40 dB HL and those showing cognitive impairment based on the Mini-Mental State Examination (MMSE) were excluded. A total of 36 individuals were assessed, of whom one was excluded due to hearing impairment and seven due to cognitive alterations. Therefore, the ic assessment was conducted with uring which data were collected on), gender (female or male), schooling level (years of study), marital status (whether or not they have a partner) and occupation (student, worker or housewife).

Mini-Mental State Examination

The participants' cognitive function was assessed using the Mini-Mental State Examination (MMSE), a neuropsychological test for screening cognitive decline that was developed by Folstein et al.[14] and translated and adapted for the Brazilian context by Bertolucci et al.[15]. The MMSE cutoff scores are set as follows: 13 points or less for illiterate individuals; 18 points for individuals with low schooling (one to eight years of studies); and 26 points for individuals with high schooling levels (more than eight years of study). The subjects whose scores fell below the cutoff for their schooling level were excluded from the sample.

Mini Sleep Questionnaire

The Mini Sleep Questionnaire assesses the frequency of sleep-related complaints based on two factors: difficulty falling asleep and overall sleep quality. The test has ten questions, each with seven answer options. The total score ranges from 10 to 70 points, with higher values indicating poorer sleep quality. The individuals were categorized into two groups: good sleep (up to 24 points) and altered sleep (25 points or more)[16,17].

Long-latency auditory evoked potential (P300)

The P300 auditory evoked potential was obtained using a Neurosoft instrument, model Neuron-Spectrum-4/EPM. To ensure precise electrode placement on the head, an elastic textile cap from MCScap was used, secured with a chin strap. The cap size was determined based on head circumference. A set of sintered Ag/AgCl electrodes with individual connectors was used, model MCScap-E.

Before attaching the electrodes, the areas where they would be placed were cleaned with alcohol and gauze. An abrasive gel was also applied to prepare the skin, with the aim of reducing skin impedance and, consequently, improving the test results. The electrodes were then positioned and the conductive gel was applied to improve conductivity and ease wave transmission

Data analysis

The data collected were entered into an Excel spreadsheet, with the participants' names replaced by numbers to ensure anonymity. The data were then exported to the Statistical Package for the Social Sciences (SPSS), version 21.0, where they were analyzed. Descriptive statistics were performed to describe the sample profile, including position and dispersion measures (mean, standard deviation, minimum and maximum values and median) for the continuous variables. Frequency tables with absolute (n) and percentage (%) values were created for the categorical variables.

Two study groups were defined based on the Mini Sleep Questionnaire results: good sleep group and altered sleep group. The groups were compared based on age, schooling level, and P300 wave latency and amplitude. Given the sample size, the Mann-Whitney test was used, preliminarily assuming data non-normality. In addition, the correlation between the data was assessed using Spearman's rank correlation coefficient. The significance level for the statistical tests was set at 5% ($p \leq 0.05$).

Characteristics	Frequency (%)	Minimum	Maximum	Mean	SD
Gender					
Female	23 (82.1)	-	-	-	-
Male	5 (17.9)	-	-	-	-
Age	-	60	87	66.75	6.29
Schooling (years)	-	2	25	11.71	6.55
0–4	8 (28.6)	-	-	-	-
5–8	2 (7.1)	-	-	-	-
>8	18 (64.3)	-	-	-	-
Occupation					
Worker/Student	7 (25.0)	-	-	-	-
Retired/Housewife	21 (75.0)	-	-	-	-
Has a partner					
Yes	16 (57.1)	-	-	-	-
No	12 (42.9)	-	-	-	-

	N (%)	Mean	SD	N (%)	Mean	SD	p-value
Female	12 (80)	-	-	11 (84)	-	-	-
Male	3 (20)	-	-	2 (15)	-	-	-
Age	15	65.80	4.39	13	66.76	7.20	0.92
Schooling	15	10.73	7.22	13	12.85	5.77	0.21
P300 latency							
Cz	14	364.85	43.97	11	364.63	74.25	0.70
Fz	13	361.76	46.02	11	355.27	57.46	0.68
Pz	13	371.84	40.48	11	352.54	61.21	0.13
P300 amplitude							
Cz	14	8.67	4.23	11	5.12	4.04	0.07*
Fz	13	8, 94	3.06	11	5.15	3.41	0.00*
Pz	13	12.51	4.60	11	7.68	3.69	0.01*

Note:*p<0.05.

Table 3. Correlation between the “sleep quality” and “age” variables, as well as “P300 wave amplitude and latency”.

		Age	Schooling	Latency			Range		
				Cz	Fz	Pz	Cz	Fz	Pz
Sleep quality	Correlation	-0.02	0.30	-0.08	-0.07	-0.32	-0.29	-0.46	-0.30
	p-value	0.88	0.117	0.69	0.73	0.12	0.15	0.02*	0.15
	N	28	28	25	24	24	25	24	24

DISCUSSION

Sleep is essential for regulating brain activity and maintaining cognitive functions, particularly in older adults, who are at greater risk of cognitive decline. During sleep, critical processes such as synaptic activity modulation and memory consolidation take place, both of which are vital for learning and reasoning[20]. Additionally, REM sleep is essential to reorganize synaptic connections, integrating new information with prior knowledge. This process is vital for brain plasticity and for the formation of long-term memories[21]. Feld and Born[22] suggest that memory consolidation during sleep not only stores new memories but also reorganizes and integrates them with prior knowledge, which in turn enhances cognitive activities such as creativity and problem-solving. Supporting these ideas, the current study showed that sleep quality is related to cognitive potential in older adults.

Event-related potentials (ERPs) represent a crucial tool for understanding the mechanisms of information processing and cognitive functions in the brain. The P300 brain wave is a component assessed through amplitude measures that reflect task-related attentional resources and latency measures, which are characterized by the reaction time to a given stimulus[23,24]. If repeatedly presented, a standard stimulus forms a well-established memory in the brain. In contrast, a target stimulus, which appears less frequently, tends to have a less consolidated memory. When a target stimulus is detected, the memory needs to be reviewed and updated. For this update to take place, attentional resources are required to identify the target stimulus and maintain the standard stimulus memory active[23]. When reacting to the target stimulus, cortical activity is recorded through P300 wave composition, which takes place approximately 300 ms after perception of the stimulus.

In the current study, cognitive potential was assessed using the P300 electrophysiological test, where yze the long-latency auditory s the precise cortical location s elicited remains uncertain[24]. und in P300 wave latency between the good sleep and altered sleep groups. However, it was observed that the altered sleep group presented a lower wave amplitude when compared to the good sleep group, particularly in the Fz and Pz channels. This result suggests that older individuals with altered sleep quality have fewer attentional resources available for task performance, thus exhibiting poorer cognitive performance as measured by P300[25].

The results of the current study partially corroborate the literature, which associates not only P300 amplitude with sleep quality but also latency, emphasizing how alterations in sleep patterns can exert impacts on cognitive performance. Reflecting the time required for information processing, P300 latency tends to be longer in individuals with sleep deprivation, leading to reduced cognitive efficiency, such as sustained attention and working memory, particularly in aged populations. Associated with attention allocation and cognitive involvement, amplitude is generally reduced in individuals with insufficient or fragmented sleep, reflecting a diminished capacity to respond to external stimuli and deficits in executive functions such as inhibitory control and cognitive flexibility[26,27].

In a study conducted with a young population, individuals with less sleep or poorer sleep quality had greater latency and reduced P300 amplitude, indicating that sleep quality exerts significant impacts on cognitive performance[27].

The study by Pavarini et al.[11] supports the idea that cognitive functions, which can be exacerbated by poor sleep quality, are reflected in the P300 measurements. These findings are consistent with the literature, which demonstrates that both acute and chronic sleep deprivation affect cognitive functions.

Acute sleep deprivation takes place when an individual stays awake for more than 24 hours, leading to immediate declines in attention and mental functioning, along with a strong feeling of sleepiness. On the other hand, chronic sleep deprivation, which makes up the basis of our study, occurs when a person consistently sleeps less than necessary over several days or weeks, resulting in gradual attention and cognitive performance declines, although the subjective feeling of sleepiness may not be as intense as in acute sleep deprivation cases[26]. Nonetheless, sleep deprivation not as difficulty focusing on specific stimuli and maintaining attention over time[28].

The hypothalamus, brainstem, thalamus, cerebral cortex and pineal gland are Central Nervous System areas that work together to regulate sleep. The subcortical hypothalamus interacts with multiple brain regions (including the prefrontal cortex) through intricate neural circuits and release of hormones. These interactions exert an influence on sleep patterns. Located in the frontal part of the brain, the prefrontal cortex plays a crucial role in cognition, planning, decision-making and social behavior, and is strongly influenced by sleep quality[29]. This finding may explain the correlation observed in the P300 wave amplitude, particularly in the frontal (Fz) channel, in individuals with poorer sleep quality.

A number of studies reinforce the link between poor sleep quality and decline in cognitive functions. In the study by Yaman et al.[30], patients with idiopathic hypersomnia showed altered sleep quality and a significant reduction in the P300 wave amplitude, reflecting deficits in attentional resources and poorer cognitive performance. These results highlight the direct relationship between poor sleep quality and decline in cognitive functions, particularly regarding sustained attention. In the study by Ray et al.[31], 24 sleep deprivation hours led to a significant decrease in the P300 wave amplitude, indicating reduced attentional resources and impaired cognitive functions. In the article by Nada et al.[13], the authors investigated the use of electrophysiological tests (including P300) to detect cognitive impairment in patients with obstructive sleep apnea-hypopnea syndrome (OSAHS). The findings revealed that, in addition to prolonged latency, the P300 wave amplitude was also significantly reduced in these patients. This decrease in P300 amplitude indicates a reduced ability to allocate attentional resources, suggesting that poor sleep quality in OSAHS patients negatively affects cognitive function.

In the current study, it is important to note that there was no median difference between the groups regarding the participants' age and schooling, as both groups were comprised by aged individuals with high schooling levels. It is well established that age and education exert significant impacts on cognitive processes, particularly in relation to cognitive reserve in older adults[11]. Thus, having groups with equivalent age and schooling levels was a favorable factor for the analysis, as it allowed for a more accurate comparison of the P300 cognitive variable between the groups, minimizing the potential influence of age and schooling on the results.

iable was not studied due to lack
tion in the sample, which can be
limitation. Future studies should
e sample size and including a more
balanced gender distribution.

In addition, it is important to note that the convenience sample comprised by 28 participants may have limited the ability to produce more robust results. These factors should be considered when interpreting the findings, as the sample is not representative of the broader aged population. Furthermore, the exclusive use of the Mini Sleep Questionnaire, which primarily assesses insomnia and hypersomnia, fails to record all aspects inherent to sleep quality. For future studies, it is recommended that more comprehensive tools be used to assess sleep quality, as they might provide a clearer understanding and greater precision regarding the relationship between sleep and cognition in aged populations.

The results of this study indicate a relationship between poor sleep quality and cognitive impairment in the elderly. Considering the effects of sleep disorders on cognitive function, it is important to implement future research should adopt longitudinal approaches to assess the impact of changes in sleep quality over time on cognition.

Some variations in the P300 latency and amplitude values between this study and others may be due to factors such as differences in equipment, participants' attention, age, time of day when the test was conducted and stimulus counting method used, among others.

AUTHORS' CONTRIBUTIONS

Conceptualization: LPCG, AJLB; Data curation: ECSJ; Formal analysis: LPCG, ECSJ, AJLB; Investigation: LPCG, ECSJ, AJLB; Methodology: LPCG, ECSJ, AJLB; Project administration: LPCG; Supervision: LPCG; Writing – original draft: LPCG, ECSJ, AJLB.